

GMX · SECURITY REVIEW

GMX

LayerZeroProvider Routing.

A security review of LayerZeroProvider Routing.

CLIENT	PROJECT	AUDITORS	AUDIT WINDOW	FINAL REPORT
GMX	LayerZeroProvider Routing	Guardian	December 20 – December 20, 2025	December 29, 2025

§ 00

Contents.

10 pages — overview, confidence ranking, scope, 1 finding, disclaimer, and about Guardian.

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§ 01

Overview.

Guardian reviewed LayerZeroProvider Routing across a single review round. The final report includes 1 Medium finding.

AUDIT FIRM	METHODOLOGY
Guardian	Smart contract · Manual · Main Review
CLIENT	
GMX	
PROJECT	
LayerZeroProvider Routing	
PREPARED BY	
Guardian	
AUDIT WINDOW	
December 20 – December 20, 2025	
FINAL REPORT	
December 29, 2025	

1.1 Confidence Ranking

5 / 5

Very High Confidence

Guardian assigns a Confidence Ranking of 5 out of 5, combining finding severity, remediation outcome, and the reviewer’s assessment of latent risk. The full scale is on the next page.

1	2	3	4	5
VERY LOW CONFIDENCE	LOW CONFIDENCE	MODERATE CONFIDENCE	HIGH CONFIDENCE	VERY HIGH CONFIDENCE

1.2 Findings at a Glance

SEVERITY	TOTAL	PENDING	ACK	PARTIAL	RESOLVED
red circle Critical	0	0	0	0	0
orange circle High	0	0	0	0	0
yellow circle Medium	1	0	1	0	0
blue circle Low	0	0	0	0	0
grey circle Informational	0	0	0	0	0
All severities	1	0	1	0	0

§ 02

Confidence Ranking.

Guardian assigns a project-level confidence score using the scale below. The ranking combines finding severity, remediation outcome, and the reviewer's assessment of latent risk.

CONFIDENCE RANKING	DEFINITION AND RECOMMENDATION	RISK PROFILE
5 Very High Confidence ASSIGNED	Codebase is mature, clean, and secure. No High or Critical vulnerabilities were found. Follows modern best practices with high test coverage and thoughtful design. Recommendation: code is highly secure at time of audit.	0 High/Critical findings and few Low/Medium severity findings.
4 High Confidence	Code is clean, well-structured, and adheres to best practices. Design patterns are sound and test coverage is strong. Recommendation: suitable for deployment after remediations; consider periodic review with changes.	0-1 High/Critical findings per engagement week and little to no Medium severity findings.
3 Moderate Confidence	Code is functional but shows gaps in defensive programming or test coverage. Recommendation: remediate the findings and re-review the changes before deployment.	Some High/Critical findings, with various Medium severity findings.
2 Low Confidence	Recurring issues suggest systemic gaps rather than isolated mistakes. Recommendation: remediate, then commission a follow-up review at a frozen commit.	Over 6 High and Critical findings across the engagement.
1 Very Low Confidence	The volume and severity of findings indicate the codebase is not ready for production. Recommendation: substantial rework, then a full re-review.	Over 12 High and Critical findings across the engagement.

§ 03

Scope & Classifications.

Guardian reviewed LayerZeroProvider Routing.

3.1 Severity Matrix

	IMP · HIGH	IMP · MED	IMP · LOW
LIK · HIGH	Critical	High	Medium
LIK · MED	High	Medium	Low
LIK · LOW	Medium	Low	Low

IMPACT · HIGH

Significant loss of assets, harm to a group of users, or core protocol functionality is disrupted.

IMPACT · MEDIUM

Small amount of funds can be lost, or ancillary functionality is affected. Users may experience reduced or delayed receipt.

IMPACT · LOW

Unexpected behavior in some functionality that is notable but does not meet the criteria for higher severity.

LIKELIHOOD · HIGH

Possible with reasonable assumptions that mimic on-chain conditions; cost is low compared to amount gained.

LIKELIHOOD · MEDIUM

Possible only in uncommon cases or requires a large amount of capital relative to disruption caused.

LIKELIHOOD · LOW

Unlikely to ever occur in production.

Files in Scope.

FILE	NSLOC	TOTAL
No files were recorded in scope.		

§ 04

Findings.

Grouped by round. Visible IDs reset by severity within each round, so a finding is named the same way in the report as it was in the channel it was written in.

ID	TITLE	CATEGORY	SEVERITY	STATUS	PAGE
MAIN REVIEW					
M-01	Messages Routed With New Data Trapped	Logical Error	MEDIUM	ACKNOWLEDGED	p. 8

MEDIUM **M-01**

ACKNOWLEDGED

Messages Routed With New Data Trapped

CATEGORY	Logical Error
LOCATION	https://arbiscan.io/address/0x7129ea01f0826c705d6f7ab01cf3c06bb83e9397#code
REVIEW	Main Review

DESCRIPTION

The diff between the old LayerZeroProvider that has been re-enabled to rescue stuck messages and the most up-to-date LayerZeroProvider is as follows:

[LayerZeroProvider Diff](<https://www.diffchecker.com/1iLL76Td/>)

In the new LayerZeroProvider the `data` payload now has a `uint256` type which is destructured from it as a second value.

The data bytes are now structured like so:

```
(ActionType, uint256, bytes)
```

If messages using this new `data` payload structure with the `uint256` as the second argument are routed to this old LayerZeroProvider contract, then the messages will be trapped forever as the provider contract is not upgradeable and the decoding of the data object as follows:

```
(ActionType actionTypes, bytes memory actionData) = abi.decode(data, (ActionType, bytes));
```

Inside the `if (srcChainId != 0 && data.length != 0) {` case will revert.

RECOMMENDATION

Ensure that messages using the new `data` format with `uint256` as the second parameter are not routed to the old LayerZeroProvider which has now been re-enabled.

RESOLUTION

Status: Acknowledged.

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Disclaimer.

This report is not, nor should be considered, an "endorsement" or "disapproval" of any particular project or team. This report is not, nor should be considered, an indication of the economics or value of any "product" or "asset" created by any team or project that contracts Guardian to perform a security assessment. This report does not provide any warranty or guarantee regarding the absolute bug-free nature of the technology analyzed, nor do they provide any indication of the technologies proprietors, business, business model or legal compliance.

This report should not be used in any way to make decisions around investment or involvement with any particular project. This report in no way provides investment advice, nor should be leveraged as investment advice of any sort. This report represents an extensive assessing process intending to help our customers increase the quality of their code while reducing the high level of risk presented by cryptographic tokens and blockchain technology.

Blockchain technology and cryptographic assets present a high level of ongoing risk. Guardian's position is that each company and individual are responsible for their own due diligence and continuous security. Guardian's goal is to help reduce the attack vectors and the high level of variance associated with utilizing new and consistently changing technologies, and in no way claims any guarantee of security or functionality of the technology we agree to analyze.

The assessment services provided by Guardian is subject to dependencies and under continuing development. You agree that your access and/or use, including but not limited to any services, reports, and materials, will be at your sole risk on an as-is, where-is, and as-available basis.

Cryptographic tokens are emergent technologies and carry with them high levels of technical risk and uncertainty. The assessment reports could include false positives, false negatives, and other unpredictable results. The services may access, and depend upon, multiple layers of third-parties.

Notice that smart contracts deployed on the blockchain are not resistant from internal/external exploit. Notice that active smart contract owner privileges constitute an elevated impact to any smart contract's safety and security. Therefore, Guardian does not guarantee the explicit security of the audited smart contract, regardless of the verdict.

§ 06

About Guardian.

Guardian is a smart contract security firm. Every engagement is worked by two independent teams competing on the same scope, each with its own tooling and methods, so no single reviewer's blind spot survives.

\$45B+DIGITAL ASSETS UNDER
REVIEW**200+**

ENGAGEMENTS COMPLETED

500+CRITICAL & HIGH FINDINGS
REPORTED

Reach us at audits@guardianaudits.com.